Cryptocurrency Market Analysis: Bitcoin, XRP, and Solana

Jeryk Coto

Spring 2026

Abstract

This project analyzes Bitcoin, XRP, and Solana using historical cryptocurrency market data. The goal of the project is to compare price trends, normalized growth, daily returns, return distributions, and machine learning prediction performance. Python was used for data cleaning, visualization, correlation analysis, and machine learning. The results show that cryptocurrency markets are highly volatile and difficult to predict, especially when analyzing short-term returns.

1 Introduction

Cryptocurrency markets have become an important part of modern financial systems. Digital assets such as Bitcoin, XRP, and Solana are traded globally and are known for rapid price movements and high volatility. Unlike traditional financial assets, cryptocurrencies are strongly affected by market demand, investor behavior, news, regulation, and economic conditions.

One major problem in cryptocurrency investing is unpredictability. Prices can increase or decrease quickly, making it difficult for investors to understand risk and predict future price movements. This project compares Bitcoin, XRP, and Solana to understand how their market behavior differs.

Bitcoin is the most established cryptocurrency and is often considered the main reference point for the crypto market. XRP is commonly associated with digital payments and financial transactions. Solana is known for fast blockchain transactions and has experienced rapid growth in recent years.

The main research question for this project is:

How do Bitcoin, XRP, and Solana compare in terms of price trends, returns, volatility, and predictability?

The objectives of this project are:

- To compare price movements of Bitcoin, XRP, and Solana.
- To analyze normalized growth to make the comparison fair.
- To examine daily returns and return distributions.
- To apply machine learning models to predict cryptocurrency returns.
- To interpret the results and explain the limitations of prediction.

2 Materials and Methods

2.1 Data

The data used in this project consists of historical cryptocurrency price data from CSV files. The three cryptocurrencies analyzed were Bitcoin, XRP, and Solana. Each dataset included daily market information.

The main variables used in the analysis were:

- Date
- Price
- Open
- High
- Low
- Volume
- Daily Return
- Volatility

The datasets were combined into one data frame. Since Solana had a shorter historical period than Bitcoin and XRP, the dataset was filtered from 2020 onward. This made the comparison more fair because all three cryptocurrencies had data during the selected time period.

2.2 Data Processing

The data was cleaned and prepared using Python in Google Colab. The date column was converted into datetime format, and the data was sorted by cryptocurrency and date. Trading volume was converted into numerical format when necessary.

Two additional features were created:

- **Daily Return:** the percentage change in price from one day to the next.
- **Volatility:** the 7-day rolling standard deviation of daily returns.

Daily return was calculated using:

$$Return_t = \frac{Price_t - Price_{t-1}}{Price_{t-1}}$$

Volatility was calculated as the rolling standard deviation of returns over a 7-day window.

2.3 Methods

The project used exploratory data analysis, data visualization, correlation analysis, and machine learning models.

The visualizations included:

- Price comparison graphs
- Normalized price comparison graph
- Daily returns graph
- Return distribution graph

The machine learning models used were:

- Linear Regression
- Ridge Regression
- Random Forest Regressor
- XGBoost Regressor

The models were evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)

3 Results and Interpretation

3.1 Price Comparison

Figure 1 shows the price movement of Bitcoin, XRP, and Solana over time. Since Bitcoin has a much higher price than XRP and Solana, the cryptocurrencies were plotted separately. This makes each trend easier to understand.

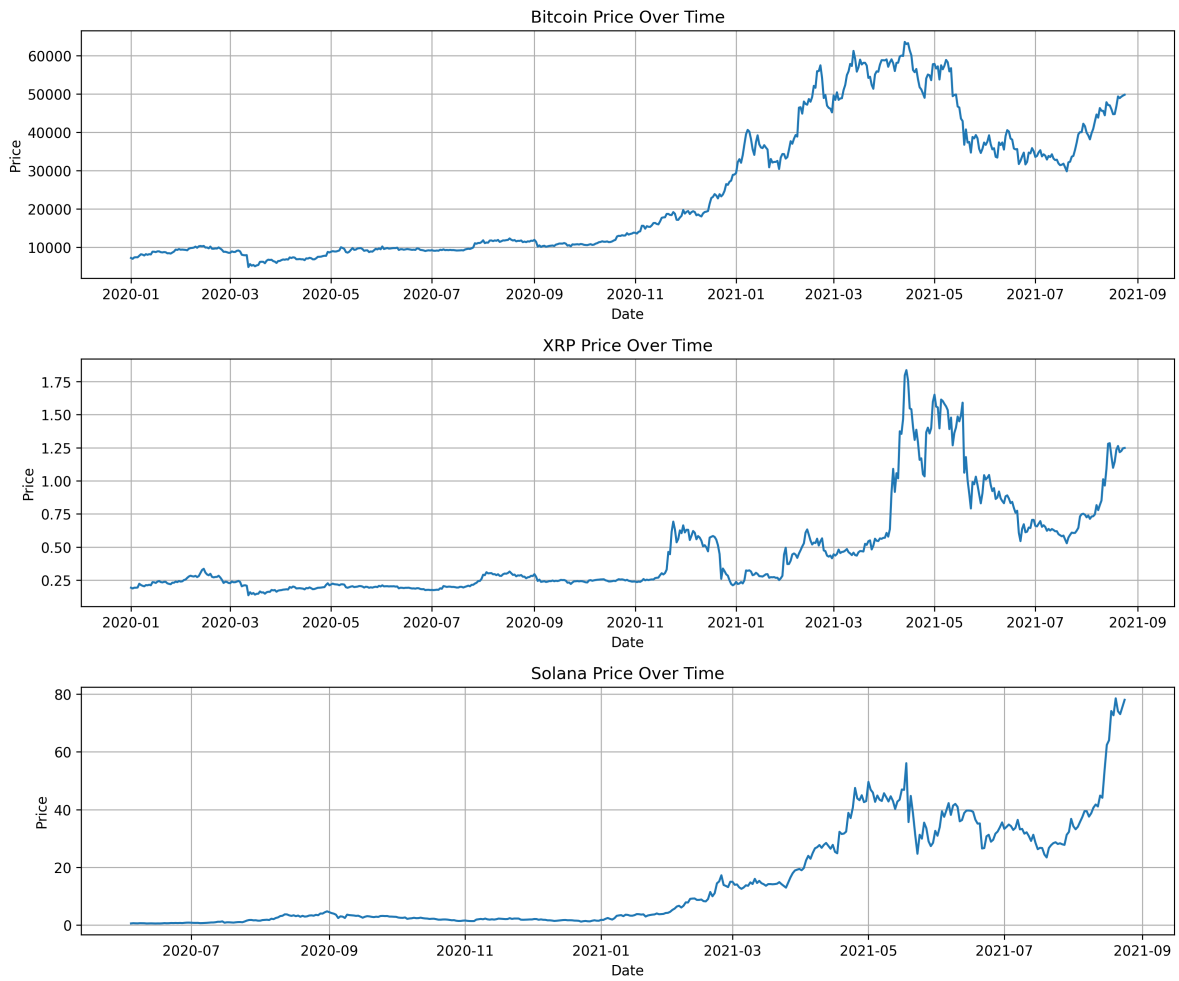

Figure 1: Price comparison of Bitcoin, XRP, and Solana.

The graph shows that Bitcoin reached the highest price level among the three cryptocurrencies. Bitcoin increased strongly from 2020 into 2021, reaching prices above $60,000 before declining and recovering later.

XRP remained at a much lower price level than Bitcoin but still showed major increases and declines. XRP experienced a sharp rise in 2021, followed by volatility.

Solana showed strong growth during the observed period. Although Solana started at a very low price, it increased quickly and reached much higher levels by 2021. This shows that Solana had strong growth potential during the selected period.

3.2 Normalized Price Comparison

Because Bitcoin, XRP, and Solana have very different price levels, a normalized price graph was created. Normalization allows the cryptocurrencies to be compared based on growth from their starting price rather than absolute price.

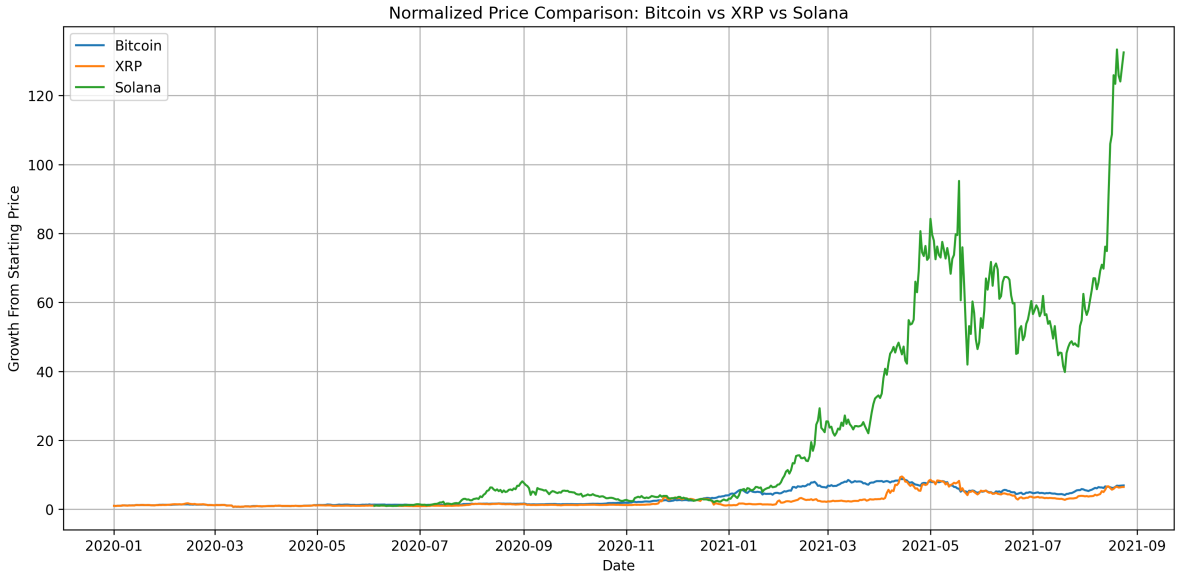

Figure 2: Normalized price comparison of Bitcoin, XRP, and Solana.

Figure 2 shows that Solana had the strongest relative growth. Solana increased more than 100 times from its starting value during the selected period. Bitcoin and XRP also increased, but their relative growth was much smaller compared to Solana.

This graph is important because it shows that the cryptocurrency with the highest price is not always the one with the highest relative growth. Bitcoin remained the most valuable in dollar price, but Solana had the strongest percentage growth.

3.3 Daily Returns

Daily returns measure how much the price changes from one day to the next. This is useful because returns show short-term movement and market risk.

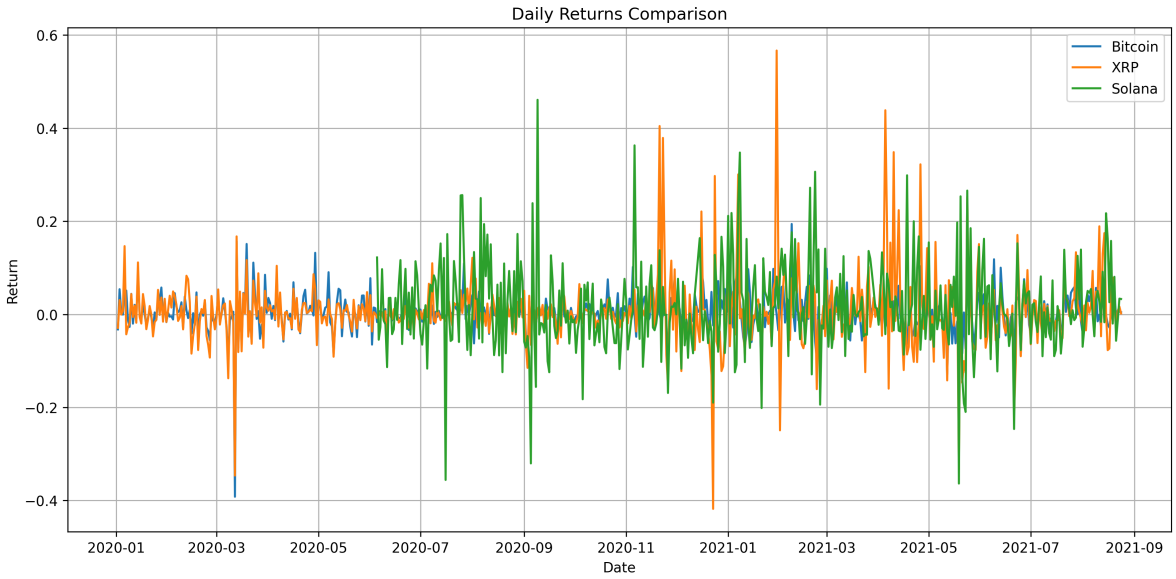

Figure 3: Daily returns comparison of Bitcoin, XRP, and Solana.

Figure 3 shows that all three cryptocurrencies experienced frequent fluctuations. Most daily returns are close to zero, but there are several large positive and negative spikes. These spikes show periods when prices changed rapidly.

Solana and XRP show larger return spikes compared to Bitcoin. This suggests that Solana and XRP were more volatile in the short term. Bitcoin also had fluctuations, but its returns appear more stable compared to the other two cryptocurrencies.

3.4 Return Distribution

The return distribution graph shows how often different return values occurred. This helps compare risk across the three cryptocurrencies.

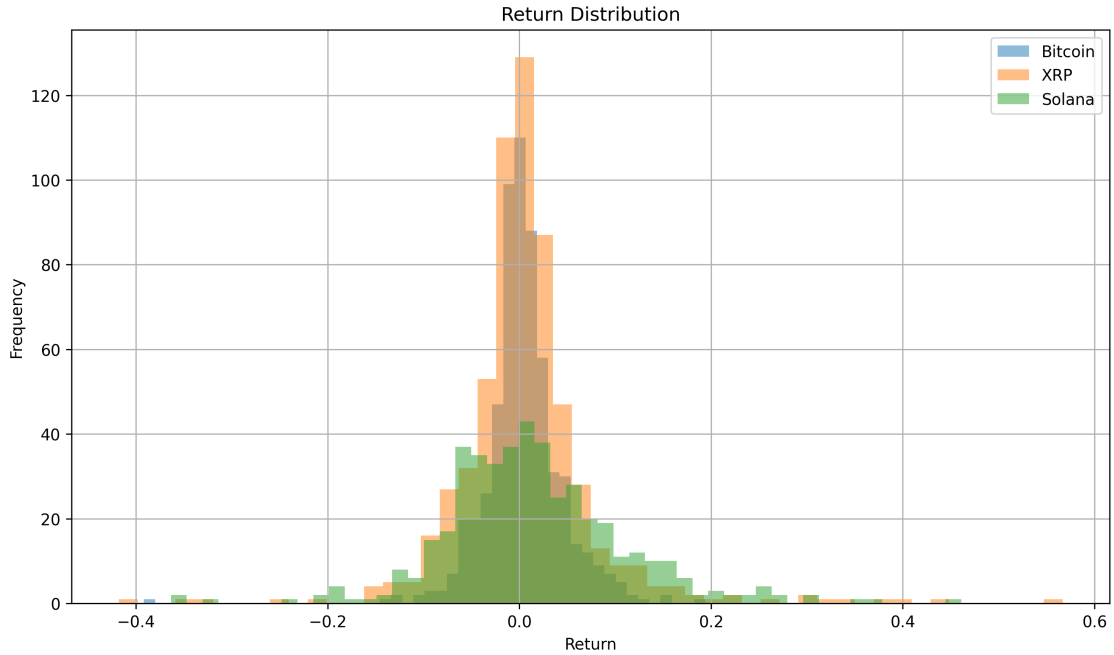

Figure 4: Return distribution of Bitcoin, XRP, and Solana.

Figure 4 shows that most returns are concentrated around zero. This means that on most days, price changes were relatively small. However, the distributions also show extreme values on both the positive and negative sides.

A wider distribution indicates greater risk and volatility. Solana shows a wider spread than Bitcoin, which suggests that Solana had larger price changes. XRP also shows large spikes, indicating periods of high volatility. Bitcoin appears more concentrated around zero, suggesting relatively lower volatility compared to XRP and Solana.

3.5 Machine Learning Results

Machine learning models were used to predict cryptocurrency returns. The target variable was daily return, and the features included market variables such as volume and volatility. Predicting returns is more realistic than predicting price because returns represent short-term market movement.

The performance of the machine learning models is shown in Table 1.

Model	MAE	RMSE	R^2
Ridge Regression	0.03455	0.04596	-0.015
Linear Regression	0.03453	0.04595	-0.014
Random Forest	0.03482	0.04598	-0.015
XGBoost	0.03600	0.04806	-0.110

Table 1: Machine learning model performance for return prediction.

The results show that all models produced similar error values. Linear Regression and Ridge Regression performed slightly better than Random Forest and XGBoost. The MAE values were around 0.034 to 0.036, and the RMSE values were around 0.045 to 0.048.

All models had negative R^2 values. This means the models did not explain the variation in cryptocurrency returns well. A negative R^2 suggests that the models performed worse than simply predicting the average return.

This result is realistic because cryptocurrency returns are highly random and difficult to predict. The results also show that more complex models do not always improve performance. Random Forest and XGBoost did not outperform simpler linear models, which suggests that the data contains a high level of noise.

4 Discussion

The analysis shows important differences between Bitcoin, XRP, and Solana. Bitcoin had the highest price level and appeared more stable compared to XRP and Solana. Solana had the strongest normalized growth, but it also showed higher volatility. XRP remained at a lower price level but experienced large short-term price movements.

The normalized price graph was especially useful because it made the comparison fair. Without normalization, Bitcoin dominates the graph because its price is much higher. After normalization, Solana clearly showed the highest relative growth.

The daily returns and return distribution graphs show that cryptocurrency markets are risky and unpredictable. Although most daily returns are close to zero, the large spikes show that prices can change quickly. This supports the idea that cryptocurrency investing involves high risk.

The machine learning results also support this conclusion. The models were not able to predict returns accurately. This suggests that short-term cryptocurrency movements may depend on outside factors such as news, investor sentiment, regulations, and market speculation.

5 Conclusion

This project analyzed Bitcoin, XRP, and Solana using historical cryptocurrency market data. The analysis compared price trends, normalized growth, daily returns, return distributions, and machine learning prediction performance.

The results show that Bitcoin is the most established cryptocurrency and had the highest price level. Solana showed the strongest relative growth, but also higher volatility and risk. XRP showed moderate performance, with several periods of large price movement.

Machine learning models were not able to accurately predict daily returns. Linear Regression and Ridge Regression performed slightly better than Random Forest and XGBoost, but all models had negative R^2 values. This shows that cryptocurrency returns are difficult to predict using only historical market variables.

Overall, this project shows that data science can help analyze market patterns, compare risk, and understand cryptocurrency behavior. However, predicting short-term cryptocurrency returns remains difficult because the market is volatile, noisy, and influenced by many external factors.

6 References

- Kaggle Cryptocurrency Historical Price Dataset.
- Scikit-learn Documentation.
- XGBoost Documentation.
- Python pandas Documentation.
- Matplotlib Documentation.
- Seaborn Documentation.